3-2:

$D = Z$

$C = Y \oplus Z$

$B = X \oplus (Y+Z)$

$A = W \oplus (X+Y+Z)$

W	X	Y	Z	A	B	C	D
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	0	0	0
1	0	0	1	0	1	1	1
1	0	1	0	0	1	1	0
1	0	1	1	0	1	0	1
1	1	0	0	0	1	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	0	1	0
1	1	1	1	0	0	0	1

3-6:

$X=0, Y=0$: $D_3 = \overline{A_3}$, $D_2 = \overline{A_2}$, $D_1 = \overline{A_1}$, $D_0 = \overline{A_0}$

$X=0, Y=1$: $D_3 = \overline{B_3}$, $D_2 = \overline{B_2}$, $D_1 = \overline{B_1}$, $D_0 = \overline{B_0}$

$X=1, Y=0$: $D_3 = \overline{C_3}$, $D_2 = \overline{C_2}$, $D_1 = \overline{C_1}$, $D_0 = \overline{C_0}$

$X=1, Y=1$: $D_3 = \overline{E_3}$, $D_2 = \overline{E_2}$, $D_1 = \overline{E_1}$, $D_0 = \overline{E_0}$

3-8:

(1). 当 $X_2X_1X_0 = Z_2Z_1Z_0$ 时 输出 0

(2) 当 $X_2X_1X_0 \neq Z_2Z_1Z_0$ 时 输出 1

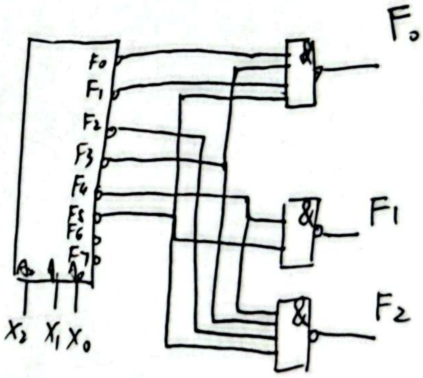
3-12

000 →	001
001	001
010	100
011	101
100	110
101	111
110	000
111	000

$F_0 = \sum m(0,1,3,5)$

$F_1 = \sum m(4,5)$

$F_2 = \sum m(2,3,4,5)$



3-3:

(a). $F = \sum m(0,4,5,6,8,10,12,15)$

(b) $F = \sum m(0,4,5,6,8,10,12,15)$

0000
~~0001~~
0100, 0101
0110
1000
1010
1100
1111

3-4:

$G_0=0, G_1=0$: $F = A$.

$G_0=0, G_1=1$:
0100 : 0
0101 : 1
0110 : 1
0111 : 0
$F = A \oplus B$

$G_0=1, G_1=0$: $A \cdot B$.

$G_0=1, G_1=1$: $A+B$

1000 : 0
1001 : 0
1010 : 0
1011 : 1

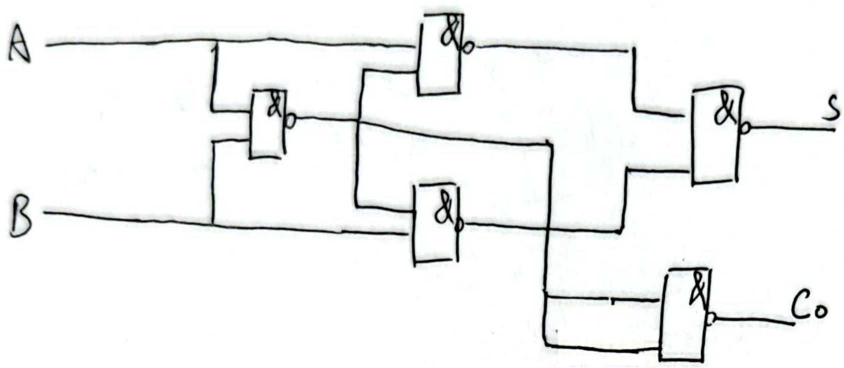
1100 : 0
1101 : 1
1110 : 1
1111 : 1

3-18:

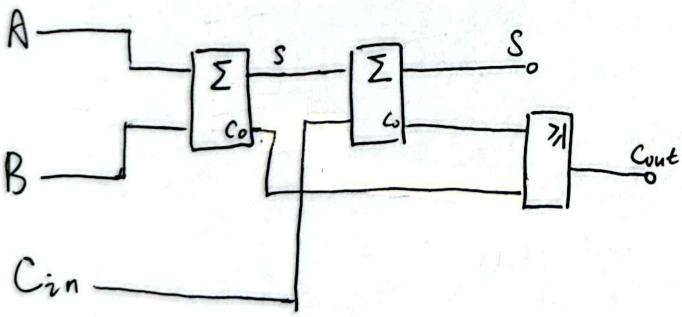
A	B	S	C_o
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$S = A \oplus B = A\overline{B} + \overline{A}B = A\cdot(\overline{AB}) + B\cdot(\overline{AB}) = \overline{(\overline{A\cdot(\overline{AB})})\cdot(\overline{B\cdot(\overline{AB})})}$$

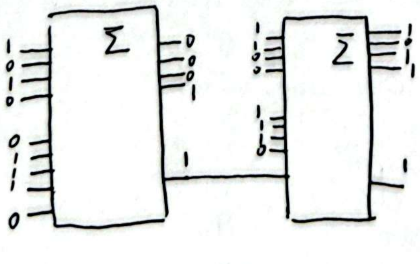
$$C_o = A\cdot B = \overline{\overline{AB}}$$



3.19



3-21



$$\begin{array}{r} 11001010 \\ +\ 11100111 \\ \hline 110110001 \end{array}$$

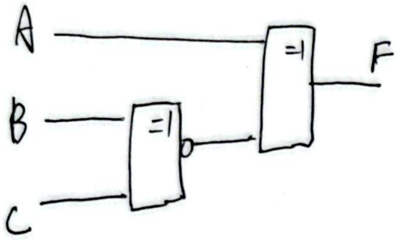
结果: 1'1011'0001

3-22.

$F=\sum m(0,3,5,6)$

A \ BC	00	01	11	10
0	1	0	1	0
1	0	1	0	1

$W_{OD} = A\oplus(B\odot C)$.



3-24

0011	3
0111	7
1001	9
1011	11
0100	4
0101	5
1100	12
1101	13
0010	2
0110	6
1010	10
1110	14

$F=\bar{A}CD+A\bar{B}D+B\bar{C}+C\bar{D}$

$F=\sum m(2,3,4,5,6,7,9,10,11,12,13,14)$

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	1	1
11	1	1	0	1
10	0	1	1	1

$\bar{A}C+A\bar{C}D+\bar{B}C+B\bar{C}+C\bar{D}$

变量A: $\bar{A}C+A\bar{C}D$.

需满足 $C=1$ 且 $\bar{C}D=1$.

不可能发生，故无竞争-冒险.

B: $\bar{B}C+B\bar{C}$.

需 $C=1$ 且 $\bar{C}=1$

不可能.

C: $C(\overline{A}+\overline{B}\overline{D})+\bar{C}(AD+B)$

需 $\overline{A}+\overline{B}\overline{D}=1$ 且 $AD+B=1$

$A=0$, $B=1$, $D=0/1$

$A=1$, $B=0$, $D=1$

$A=1$, $B=1$, $D=0$

D: 不可能